THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

2. $\dfrac{3^6 \times 3^4}{3^5}$ equals: 1

(a) 3^{19} **(b)** 3^{15} **(c)** 3^5 **(d)** 3^7

3. The decimal value of 10110_2 is: 1

(a) 28 **(b)** 20 **(c)** 24 **(d)** 22

4. The angles of a quadrilateral are x°, $(x+10)^\circ$, $(x+20)^\circ$ and $(x+30)^\circ$. The value of x is: 1

(a) 65 **(b)** 76 **(c)** 85 **(d)** 75

5. The digit that replaces * in 7*93 so that the number is divisible by 11 is: 1

(a) 2 **(b)** 5 **(c)** 8 **(d)** 3

6. $(3x+2)(2x+5)$ equals: 1

(a) $6x^2+10$ **(b)** $6x^2+19x-10$ **(c)** $6x^2+19x+10$ **(d)** $6x^2-19x+10$

7. Two numbers are in the ratio 4 : 7 and their sum is 99. The larger number is: 1

(a) 72 **(b)** 63 **(c)** 36 **(d)** 49

8. An article is sold for ₹ 1260 at a profit of 5%. Its cost price (in ₹) is: 1

(a) 1320 **(b)** 1200 **(c)** 1080 **(d)** 1197

9. An isosceles triangle has equal sides 30 cm and base 36 cm. Its height (in cm) is: 1

(a) 23 **(b)** 25 **(c)** 24 **(d)** 12

10. If A:B = 3:4 and B:C = 8:9, then A:C is: 1

(a) 3:9 **(b)** 27:32 **(c)** 2:3 **(d)** 3:5

11. A prism has 18 edges. The number of its faces is: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

12. The mean mark of 10 students is 40 and of another 15 students is 60. The mean of all 25 students is: 1

(a) 1300 **(b)** 54 **(c)** 52 **(d)** 50

13. On expanding $3(x+4)+2x$ we get: 1

(a) $3x+14$ **(b)** $6x+12$ **(c)** $5x+4$ **(d)** $5x+12$

14. An L-shaped figure is made by removing a 8 cm by 6 cm rectangle from a 20 cm by 15 cm rectangle. Its area is: 1

(a) 108 cm² **(b)** 300 cm² **(c)** 252 cm² **(d)** 348 cm²

15. The smallest number by which 192 must be divided to get a perfect cube is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

16. The value of $\left(\frac{2}{3}\right)^{-2}$ is: 1

(a) $\frac{9}{4}$ **(b)** $-\frac{9}{4}$ **(c)** 3 **(d)** $\frac{4}{9}$

17. The binary form of 45 is: 1

(a) 101100 **(b)** 110001 **(c)** 101110 **(d)** 101101

18. The adjacent sides of a parallelogram are in the ratio 3:5 and its perimeter is 64 cm. The longer side is: 1

(a) 40 cm **(b)** 32 cm **(c)** 12 cm **(d)** 20 cm

19. The smallest digit that replaces * in 52*8 so that the number is divisible by 6 is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 5

20. $(2x - 3y)^2$ equals: 1

(a) $4x^2 + 12xy + 9y^2$ **(b)** $4x^2 - 9y^2$ **(c)** $4x^2 - 12xy + 9y^2$ **(d)** $4x^2 - 6xy + 9y^2$

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find $\sqrt{2916}$ by prime factorisation. 2

22. Find the value of $(2^3)^3 \div 2^5$. 2

23. Convert 110101_2 to decimal. 2

24. The angles of a quadrilateral are in the ratio 2:3:5:8. Find the largest angle. 2

25. Find the smallest number to be added to 8456 to make it divisible by 11. 2

26. Expand $(2x + 5y)^2$. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Find the smallest number by which 1872 must be multiplied to make it a perfect square. Find the square root of the product. 3

28. Simplify $\frac{2^5 \times 3^4}{2^3 \times 3^2}$ and find its value. 3

29. Convert 101101_2 to base 8 by first converting to decimal. 3

30. The angles of a quadrilateral are x°, $(x + 15)^\circ$, $(x + 30)^\circ$ and $(x + 45)^\circ$. Find all the angles. 3

31. Find the smallest 4-digit number divisible by both 12 and 15. 3

OR

Expand $(3x + 2)(2x - 5)$. 3

32. A mixture of 40 L contains milk and water in the ratio 7 : 1. How much water must be added to make the ratio 7 : 3? 3

OR

A shopkeeper allows successive discounts of 10% and 5% on a marked price of ₹ 4000. Find the selling price and the single equivalent discount percent. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A school has 1500 students who are to stand in rows so that the number of rows equals the number of students in each row. How many more students are needed to complete the arrangement? What is the number of rows then? 5

34. Simplify $\dfrac{2^{3} \times 3^{-1}}{2^{-2} \times 3^{1}}$ and write the answer with positive exponents. Then find its value. 5

35. Find the sum of 1101_2, 1001_2 and 110_2. Verify the result in decimal. 5

36. ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (2x + 10)^\circ$ and $\angle OBA = (3x - 5)^\circ$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

OR

Apply the Kaprekar routine to 1495 until you reach 6174. Write every step and state the number of steps. 5

37. A square plot has side $(3x + 2)$ m. A square pond of side $2x$ m is dug inside it. Write the area of the remaining land as an expression in x and find it when $x = 10$. 5

OR

12 men can complete a piece of work in 15 days. 5
(i) How many days will 18 men take?
(ii) How many men are needed to finish it in 9 days?

38. A trader sells a radio for ₹ 2100 and gains 5%. If he had sold it for ₹ 2400, what would be his gain or loss percent? 5

OR

A man walks 18 m east and then 24 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND